Worked Example 1 — Compound interest

Question: Find the amount and the compound interest when ₹12,000 is invested at 8% per annum, compounded annually, for 3 years.

Solution:

Step 1: Write the formula $A = P(1 + \frac{r}{100})^n$, where P is the principal, r the rate per cent per annum and n the number of years.

Step 2: Substitute P = 12000, r = 8 and n = 3: $A = 12000 \times (1 + \frac{8}{100})^3 = 12000 \times (1.08)^3$.

Step 3: Work out the growth factor: $(1.08)^3 = 1.08 \times 1.08 \times 1.08 = 1.259712$.

Step 4: Multiply: $A = 12000 \times 1.259712 = 15116.544$, so $A \approx$ ₹15,116.54.

Step 5: Compound interest = A − P = 15116.54 − 12000 = ₹3,116.54.

Answer: The amount is ₹15,116.54 and the compound interest is ₹3,116.54.

Worked Example 2 — Solving a quadratic equation with the formula

Question: Solve $2x^2 - 7x + 3 = 0$.

Solution:

Step 1: Compare with $ax^2 + bx + c = 0$: a = 2, b = −7, c = 3.

Step 2: Find the discriminant $D = b^2 - 4ac = (-7)^2 - 4 \times 2 \times 3 = 49 - 24 = 25$.

Step 3: Since D > 0 there are two real roots, and $\sqrt{D} = 5$.

Step 4: Use $x = \frac{-b \pm \sqrt{D}}{2a} = \frac{7 \pm 5}{4}$.

Step 5: $x = \frac{12}{4} = 3$ or $x = \frac{2}{4} = ½$.

Answer: x = 3 or x = ½.

Worked Example 3 — Simple interest

Question: Find the simple interest on ₹5,000 at 6% per annum for 4 years.

Solution:

Step 1: Write the formula SI = $\frac{\text{P} \times \text{R} \times \text{T}}{100}$.

Step 2: Substitute P = 5000, R = 6 and T = 4.

Step 3: SI = $\frac{5000 \times 6 \times 4}{100} = \frac{120000}{100} = 1200$.

Step 4: Amount = P + SI = 5000 + 1200 = 6200.

Answer: The simple interest is ₹1,200 and the amount is ₹6,200.

Worked Example 4 — Percentage increase

Question: A shirt's price rises from ₹800 to ₹920. Find the percentage increase.

Solution:

Step 1: Increase = new price − old price = 920 − 800 = 120.

Step 2: Percentage increase = $\frac{\text{increase}}{\text{original price}} \times 100$.

Step 3: $\frac{120}{800} \times 100 = 15$.

Answer: The price increased by 15%.

Worked Example 5 — Average speed

Question: A car travels 150 km at 50 km/h and then 150 km at 75 km/h. Find the average speed for the whole journey.

Solution:

Step 1: Time for the first part = 150 ÷ 50 = 3 h.

Step 2: Time for the second part = 150 ÷ 75 = 2 h.

Step 3: Total distance = 300 km and total time = 5 h.

Step 4: Average speed = total distance ÷ total time = 300 ÷ 5 = 60 km/h.

Note: the average speed is not the mean of 50 and 75, because the car spends longer at the slower speed.

Worked Example 6 — Area of a trapezium

Question: Find the area of a trapezium whose parallel sides are 10 cm and 6 cm and whose height is 4 cm.

Solution:

Step 1: Write the formula Area = ½ × (a + b) × h.

Step 2: Substitute a = 10, b = 6 and h = 4.

Step 3: Area = ½ × 16 × 4 = 32.

Answer: The area is 32 cm².